EE4047 – High Voltage Techniques I

Continuation to:

The Fundamentals of High Voltage Engineering

Week 3.2 - Thursday Lesson

Asst. Prof. Dr. M. Alparslan

16.03.2023

OUTLINE

3 **Continuation to:** **The Fundamentals of High Voltage Engineering**

High Voltage Elements

Circuit Breakers

	Openning (Breaking)			Closing (Making)			Isolation
	No load	Load	Short circuit	No load	Load	Short circuit	
Disconnector	YES	NO	NO	YES	NO	YES	YES
Ground switch	YES	NO	NO	YES	NO	YES	NO
Switch	YES	YES	NO	YES	YES	YES	YES
Contactor	YES	YES	NO	YES	YES	YES	NO
Circuit breaker	YES	YES	YES	YES	YES	YES	NO

High Voltage Elements

Circuit Breakers

They can be opened/closed manually or automatically.

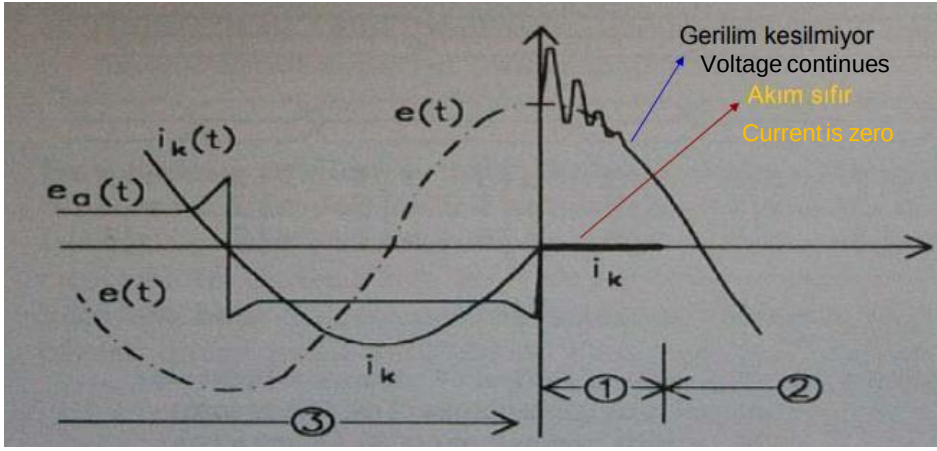
The current is interrupted; while the contact **still have voltage**.

In the figure:

1 Transient recovery voltage

2 Continuous recovery voltage

3 Breaker openning action



High Voltage Elements

Circuit Breakers

Recovery voltage depends on,

- power factor of the circuit,
- if the short circuit is symmetrical or not,
- armature reaction in generators.

High Voltage Elements

Circuit Breakers

Some label information about circuit breakers:

- U_n (kV) rated (nominal) voltage: The voltage level at which the circuit breaker can operate continuously.

- I_n **(A) rated (nominal) current: The maximum current value that can flow over the circuit breaker.**

- f (Hz) operation frequency: The frequency of the grid that circuit breaker is operating as part of.

- Breaking current (A): The current that flows through the breaker in the moment when the contacts are disconnected. It can be symmetrical or asymmetrical depending on time axis.

Rated breaking current (A): The symmetrical current broken when tripping command is given.

High Voltage Elements

Circuit Breakers

Types of circuit breakers

- Air circuit breakers
- Oil circuit breakers
    - Bulk oil circuit breakers
    - Minimum oil circuit breaker (more developed version, cheaper, easier installation, more maintenance need, not widely preferred today)
- Gas **(**SF_6**)** circuit breakers
- Vacuum circuit breakers
- HVDC circuit breakers

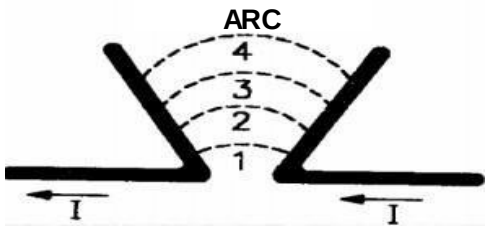
THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

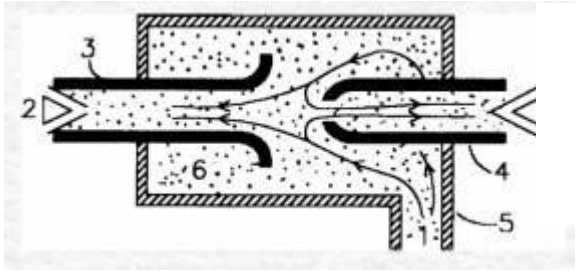
High Voltage Elements

Circuit Breakers

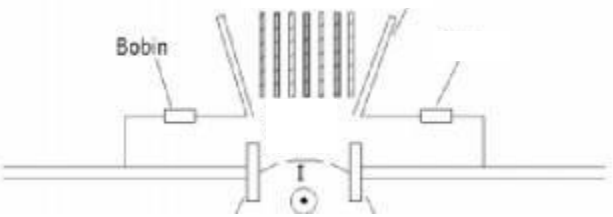
Types of circuit breakers

- Air circuit breakers





High pressure
air breaking



Arc splitting

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

High Voltage Elements

Circuit Breakers

Types of circuit breakers

- Gas **(SF_6)** circuit breakers

This gas type has dielectric strenght 2-3 times of air. It has good arc extinguishing characteristic.

It is a heavy gas, so in case of leakeges, it does not mix in air and go away. It stays at certain spaces.

In normal conditions it does not have colour, smell, taste, poisonous impact or flamability. It also has good thermal conductivity.

However, when used in circuit breakers with some other mixtures and during arcs, it can lead to poisonous cases.

Also used in gas insulated cables, gas insulated disconnectors, load disconnectors and transformer substations.

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

- **High Voltage Elements**

Circuit Breakers

Types of circuit breakers

- Vacuum circuit breakers

No arc extinguishing or cooling mechanisms are used.

Less maintenance need.

Involved less pieces, highly reliable, lower risk of facing problems.

Can even break capacitive currents and very small inductive currents.

High Voltage Elements

Circuit Breakers (Comparison of types)

	Oil	Air	SF_6 / Vacuum
Security	Can explode, burn, increasing pressure inside the tank is risky.	Can be impacted by external factors, while breaking hot and ionized gas is emitted.	No impact due to external factors, no risk of explosion or flamability.
Size	Large	Large	Small
Maintenance	Frequent, change of oil	Change of contacts, control of mechanical parts	No maintenance of breaker parts, minimal oiling for mechanical parts
Sensitivity to environmental factors	Sensitive	Sensitive	Not sensitive, breaking takes place in closed space
Recovery time	Lower breaking capability in case of increased pressure	Lower breaking capability in cae of hot air emission	Fast dielectric recovery, can break consecutively.
Durability	Medium	Medium	Excellent

High Voltage Elements

Circuit Breakers



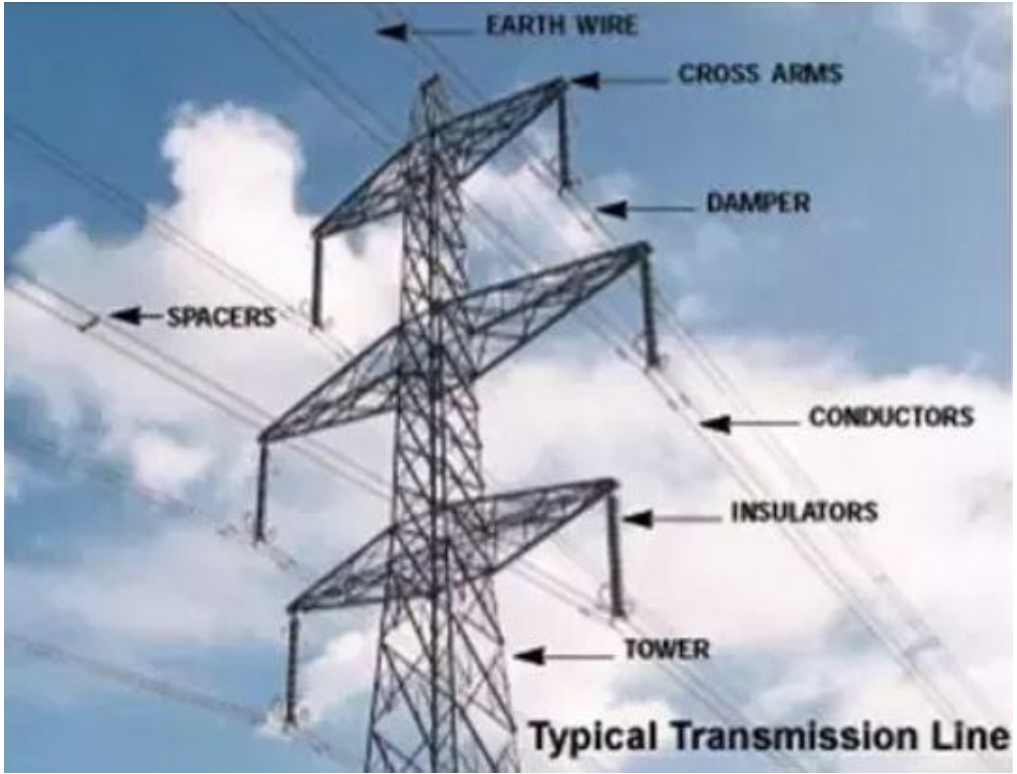
Yalıtılmış gövdeli (live tank)



Topraklı gövdeli (dead tank)

High Voltage Elements

Power Transmission Lines

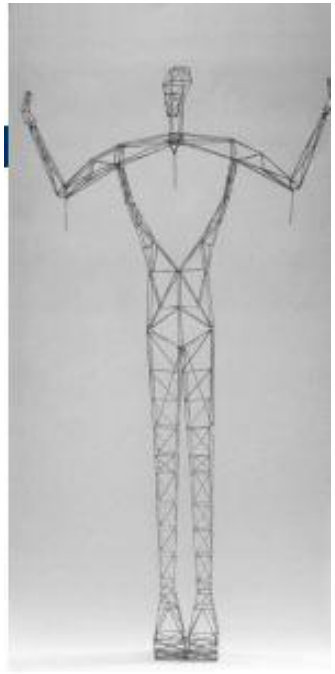


High Voltage Elements

Power Transmission Lines (Fantastic designs and applications)

Iceland





High Voltage Elements

Power Transmission Lines (Fantastic designs and applications)

Marsellie, France



High Voltage Elements

Power Transmission Lines (Fantastic designs and applications)

South Korea



High Voltage Elements

Power Transmission Lines (Fantastic designs and applications)

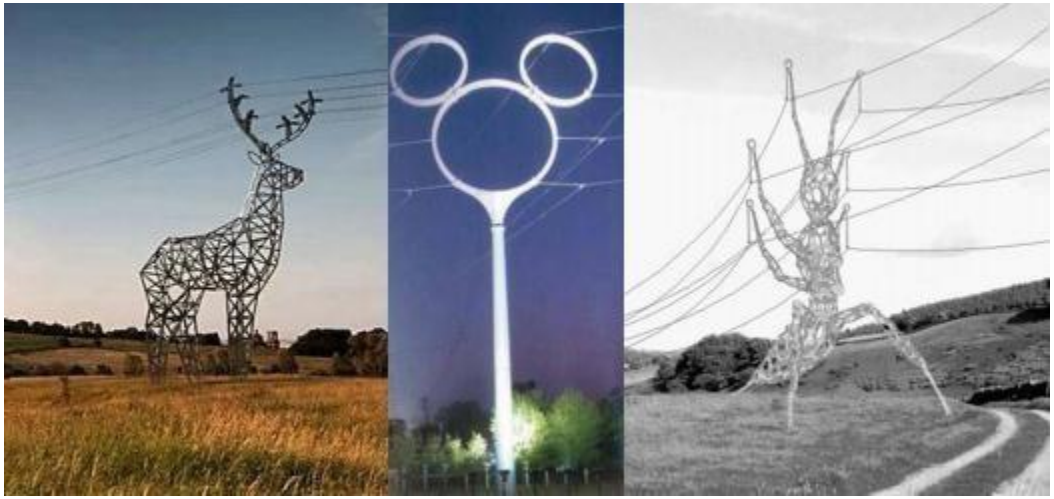
Norway and Iceland



High Voltage Elements

Power Transmission Lines (Fantastic designs and applications)

Orlando, Florida, US (near Disneyland)





High Voltage Elements

Conductor types used in transmission lines:

1. Copper
2. Aluminium
3. Aldrey
4. Cadmium Copper
5. Galvanized Steel
6. Phophor Bronze
7. Copper Steel Braid
8. Aluminium Steel Braid



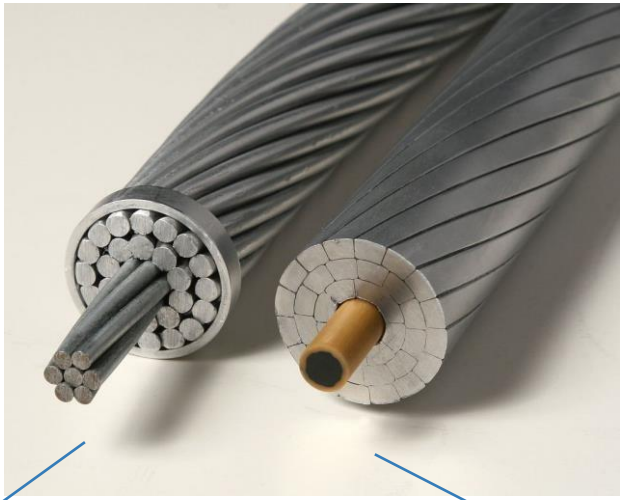
High Voltage Elements

Conductor types used in transmission lines:

1. Copper
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6. Phophor Bronze
7. Copper Steel Braid
8. Aluminium Steel Braid

High Voltage Elements

Transmission lines





Aluminium Conductor Steel Reinforced (ACSR)

Aluminium Composite Core (ACCC)

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

High Voltage Elements

Transmission lines (bundle conductor)







Spacer

High Voltage Elements

Transmission lines (3-bundled conductor)





High Voltage Elements

Transmission lines (4-bundled conductor)





High Voltage Elements

Poles, pylons classification

A. According to their structure

1. Wood pylons
2. Concrete pylons
3. Lattice steel pylons

B. According to their circuits

1. DC
2. AC
   a. Single circuit
   b. Double circuit



High Voltage Elements

Poles, pylons classification

C. According to their function

1. Carrier poles
   a. Normal carrier poles
   b. Corner carrier poles
2. Anchor poles
   a. Normal anchor poles
   b. Corner anchor poles

D. Distribution poles

E. Transformer pylons

F. Breaker carrying pylons

G. Termination pylon

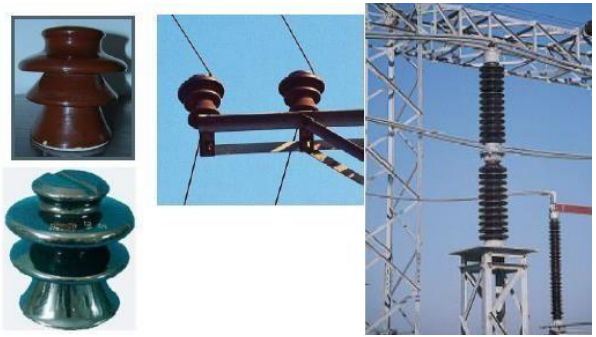
H. Branch pylons

High Voltage Elements

High voltage insulators

A. According to their structure

1. Suspension insulator
   a. Normal type
   b. Fog type
2. Strain insulator
   a. Single element strain type
   b. Multiple element strain type
3. Bushing insulators
   a. Rod type
   b. Capacitive type







High Voltage Elements

High voltage insulators

B. According to area of use

1. Indoor insulator
2. Outdoor insulator



C. According to conditions

1. According to normal weather conditions
2. According to heavy weather conditions





D. According to material

1. Porcelain insulators
2. Glass insulators
3. Epoxy resins insulators
4. Silicone insulators
5. Silicone rubber insulators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

High Voltage Elements

High voltage insulators

String insulators





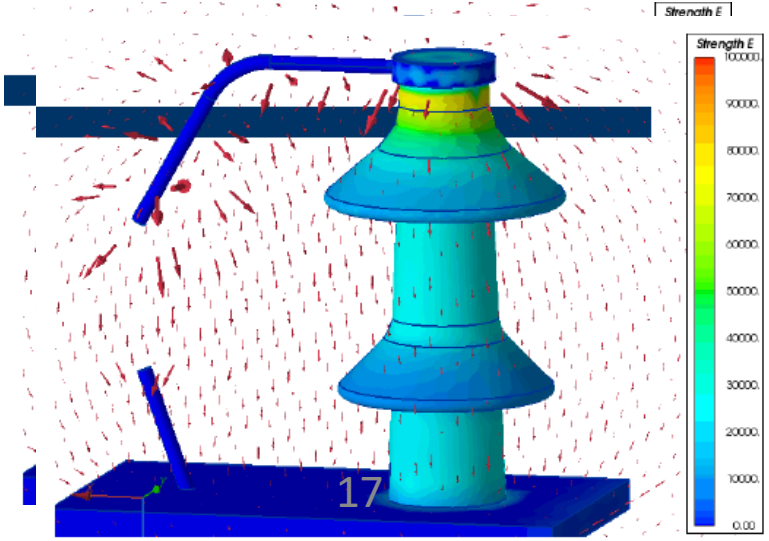
Insulator string



High Voltage Elements

High voltage insulators

Insulator electric field analysis example



High Voltage Elements

High voltage insulators

Transormer bushing insulators





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High Voltage Elements

High voltage insulators

Wall bushing insulators



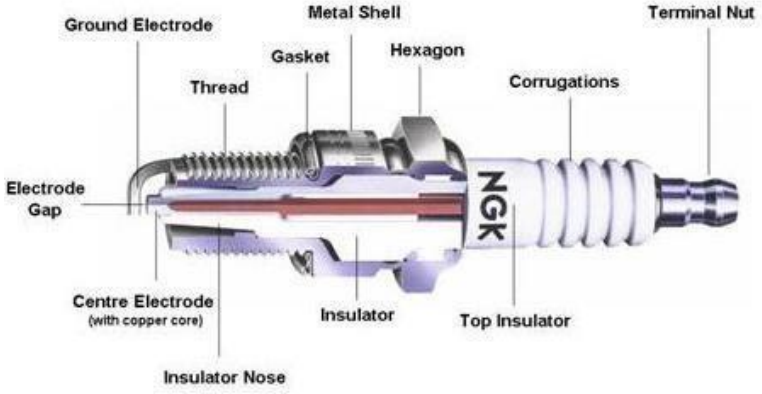


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High Voltage Elements

Spark plug

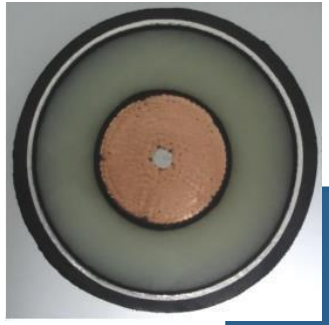




High Voltage Elements

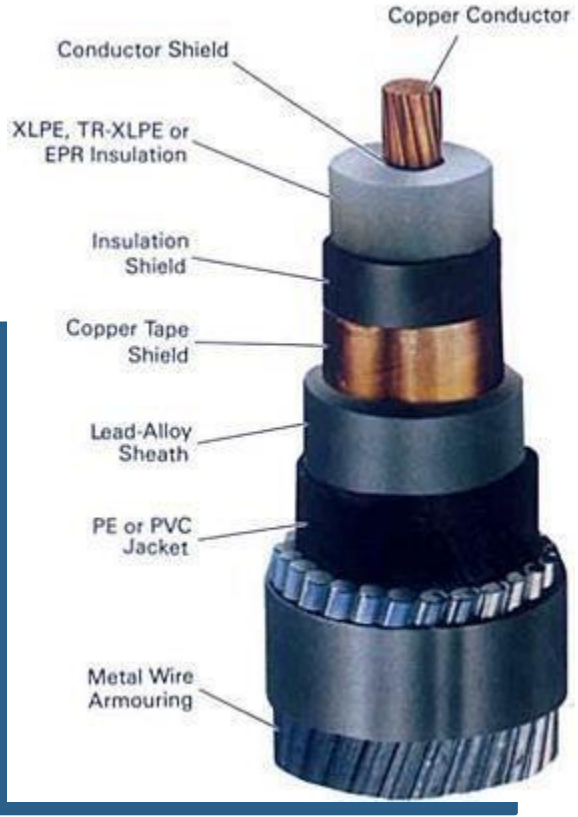
High voltage cables











High Voltage Elements

Cable terminations





High Voltage Elements

Cable joints



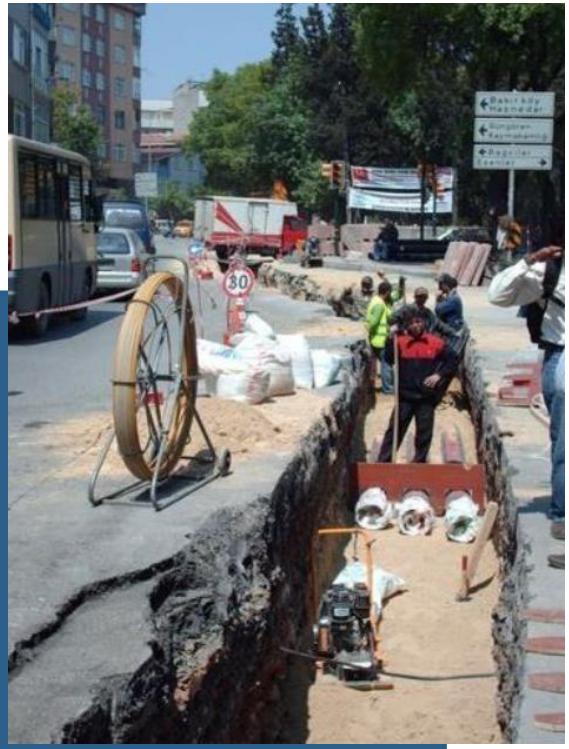


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High Voltage Elements

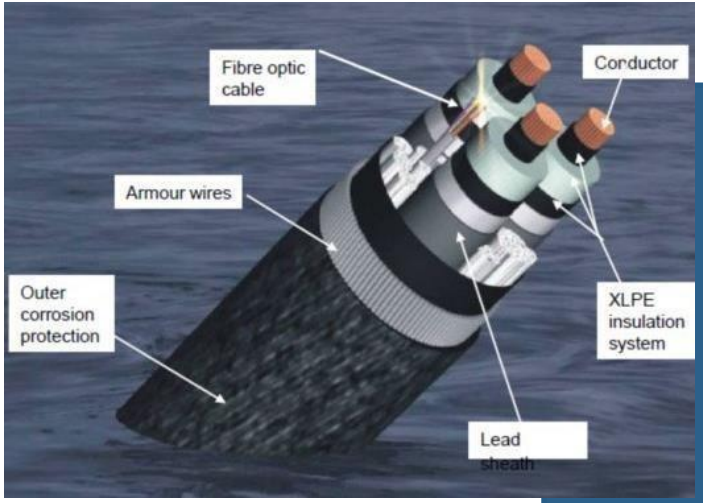
HV cables





High Voltage Elements

HV cables







High Voltage Elements

Buses





High Voltage Elements

Current and voltage transformers



Akım ölçü trafosu
Current transformer



Akım ölçü trafosu
Current transformer



Gerilim ölçü trafosu
Voltage transformer



erilim ölçü trafosu
Voltage transformer

High Voltage Elements

Voltage dividers



Capacitive voltage divider

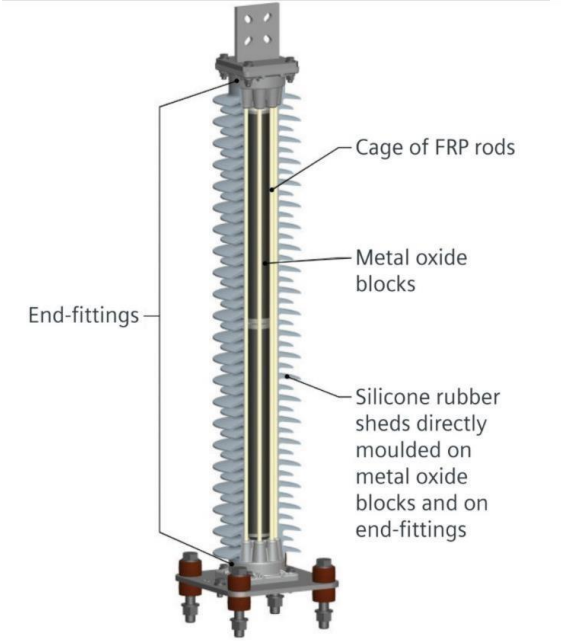


Resistive voltage divider

High Voltage Elements

Surge arrester, Lightning arrester







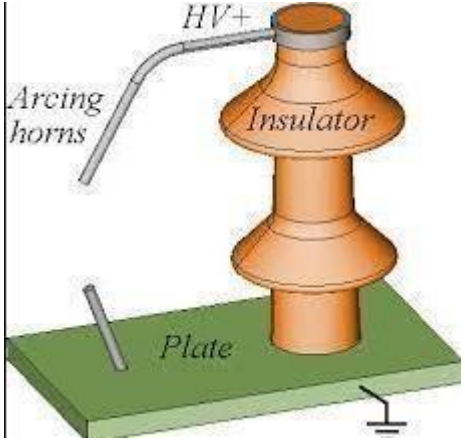


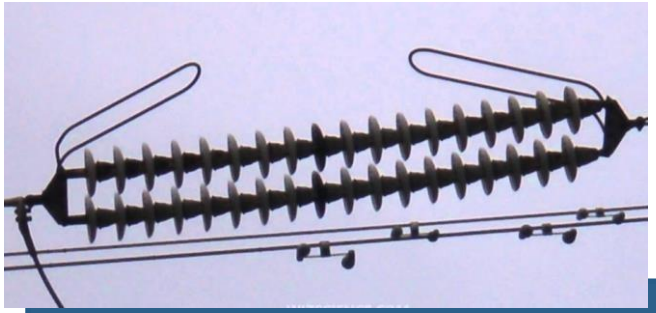


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High Voltage Elements

Arcing horns











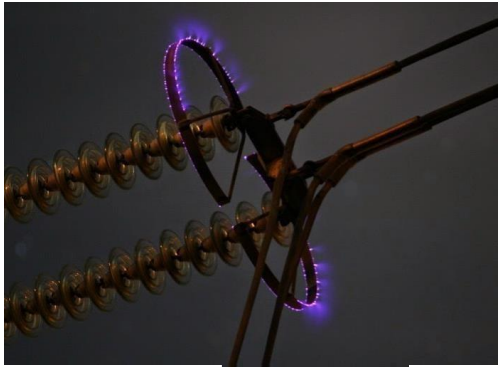
High Voltage Elements

Corona rings









3

High Voltage Elements

High voltage capacitors





3

High Voltage Elements

Transformer substations





High Voltage Elements

Gas insulated (GIS) Transformer substations (indoor)





High Voltage Elements

Concrete kiosk



High Voltage Elements

HV fuses





3

High Voltage Elements

Pole climbing equipment and earthing rod/stick





EE4047 – High Voltage Techniques I

New Topic:
Generation of High Voltage Engineerings

Week 3.2 - Thursday Lesson

Asst. Prof. Dr. M. Alparslan

16.03.2023

END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm

OUTLINE

1 Introduction

2 Direct Voltages

2.1 AC to DC conversion

2.2 Electrostatic generators

3 Alternating Voltages

3.1 Testing transformers

3.2 Series resonant circuits

4 Impulse Voltages

5 Control Systems

1 INTRODUCTION

Generally, commercially available high voltage generators are applied in routine testing laboratories. They are used for testing equipment such as transformers, bushings, cables, capacitors, switchgear and other.

The tests should confirm the efficiency and reliability of the products.

HV testing equpment is required to study the insulation behavior under all conditions, which the apparatus is likely to encounter.

The amplitudes and types of test voltages, which are always higher than the normal or rated voltages of the apparatus under test, are in general prescribed by national or international standards or recommendations.

Routine testing laboratories are also used for the development of new products. Then, even higher voltages might be necessary.

1 INTRODUCTION

Most of the HV generator circuits can be changed to increase the output voltage leves, if the original circuit was properly designed. Therefore, even the selection of routine testing equipment should always consider a future extension of the testing capabilities.

2 DIRECT VOLTAGES

In the International Standard IEC 60-1 or IEEE Standard 4-1995, the value of a direct test voltage is defined by its arithmetic mean value, $\bar{V}$, which may be derived from

$$\bar{V} = \frac{1}{T}\int_0^T V(t)\,dt$$

where T equals a certain period of time if the voltage $V(t)$ is not constant, but periodically oscillating with a frequency of $f = 1/T$

Test voltages as applied to test objects then deviate periodically from the mean value. This means that a ripple is present. The amplitude of the ripple δV is defined as half the difference between the maximum and minimum values,

$$\delta V = 0.5(V_{max} - V_{min})$$

The ripple factor is the ratio of the ripple amplitude to the arithmetic mean value, or $\delta V/\bar{V}$. For test voltages this ripple factor should not exceed 3% unless otherwise specified by the appropriate standard or be necessary for fundamental investigations.

2 DIRECT VOLTAGES

The DC voltages are generally obtained by rectifying circuits applied to AC voltages or by electrostatic generation.

2.1 AC to DC conversion

The rectification of AC to DC is the most efficient means of obtaining HVDC supplies.

Simple rectifier circuits

For a clear understanding o fall AC to DC conversion circuits, the signle-phase half-wave rectifier with volage smoothing is of basic interest. The circuit structure is given in the figure.

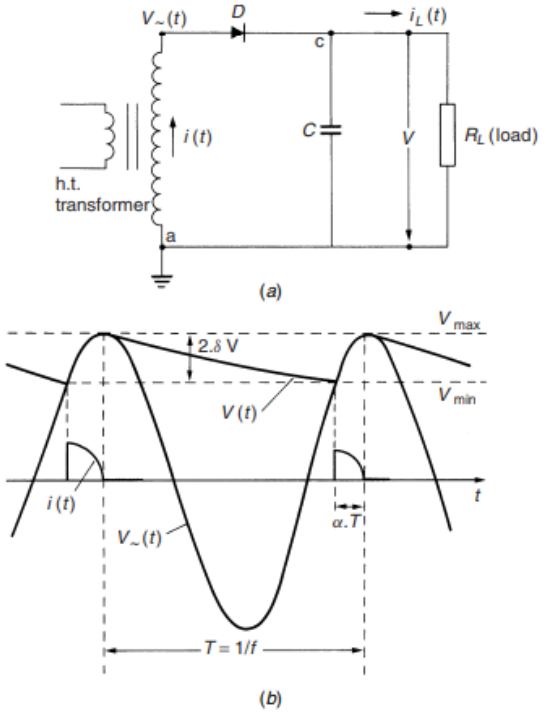


Figure 2.1 *Single-phase half-wave rectifier with reservoir capacitance C. (a) Circuit. (b) Voltages and currents with load R_L*

2 DIRECT VOLTAGES

If we neglect the leakage reactance of the transformer and the small internal impedance of the diodes during conduction, the smoothing capacitor C is charged to the maximum voltage $+V_{max}$ of the AC voltage $V_{\sim}(t)$ of the HV transfomer when D conducts.

This is the case as long as $V < V_{\sim}(t)$ for the polarity of D assumed.

If $I = 0$, the output load being zero ($R_L = \infty$), the DC voltage across C remains constant ($+V_{max}$), whereas $V_{\sim}(t)$ oscillates between $\pm V_{max}$. The diode must be dimensioned to withstand a peak reverse voltage of $2V_{max}$.

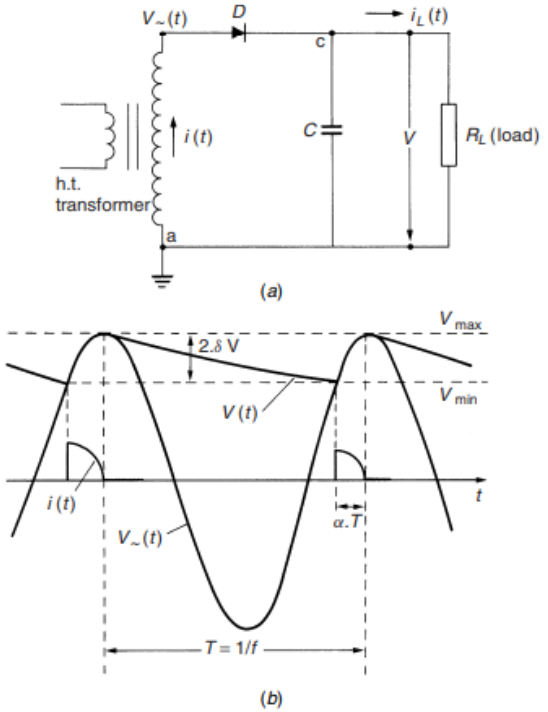


Figure 2.1 *Single-phase half-wave rectifier with reservoir capacitance C. (a) Circuit. (b) Voltages and currents with load R_L*

DIRECT VOLTAGES

The output voltage V does not remain any more constant if the circuit is loaded. During one period, $T = 1/f$ of the AC voltage a charge Q is transferred to the load R_L,

$$Q = \int_T i_L(t)dt = \frac{1}{R_L}\int_T V(t)dt = IT = \frac{I}{f}$$

I is the mean value of the DC output $i_L(t)$ and $V(t)$ the DC voltage shich includes a ripple shown in the figure.

If we introduce the ripple factor δV we may easily see that $V(t)$ varies between

$$V_{max} \geq V(t) \geq V_{min}\ V_{min} = V_{max} - 2(\delta V)$$

The charge Q is also supplied from the transformer within the short conduction time $t_c = \alpha T$

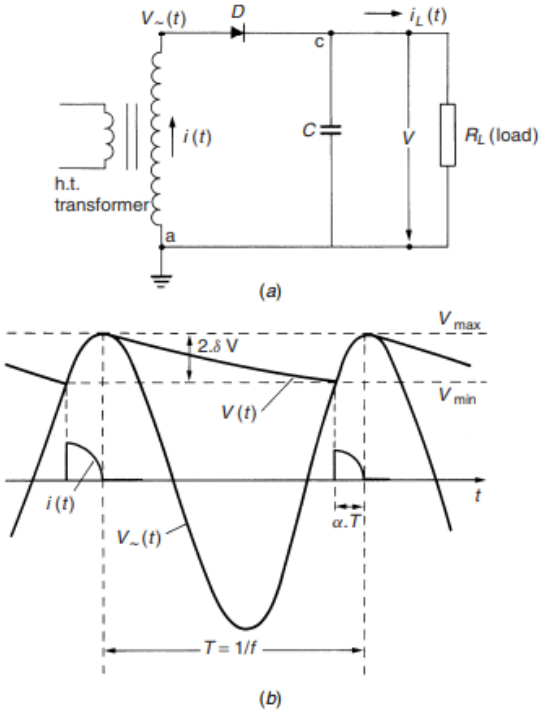


Figure 2.1 *Single-phase half-wave rectifier with reservoir capacitance C. (a) Circuit. (b) Voltages and currents with load R_L*

2 DIRECT VOLTAGES

The charge Q is also supplied from the transformer within the short conduction time $t_c = \alpha T$ of the diod D during each cycle.

Therefore Q equals to

$$Q = \int_{\alpha T} i(t)dt = \int_{T} i_L(t)dt$$

As $\alpha T \ll T$, the transfomer and diode current $i(t)$ is pulsed as shown in the figure and is of much bigger amplitudes than the direct current $i_L \cong I$.

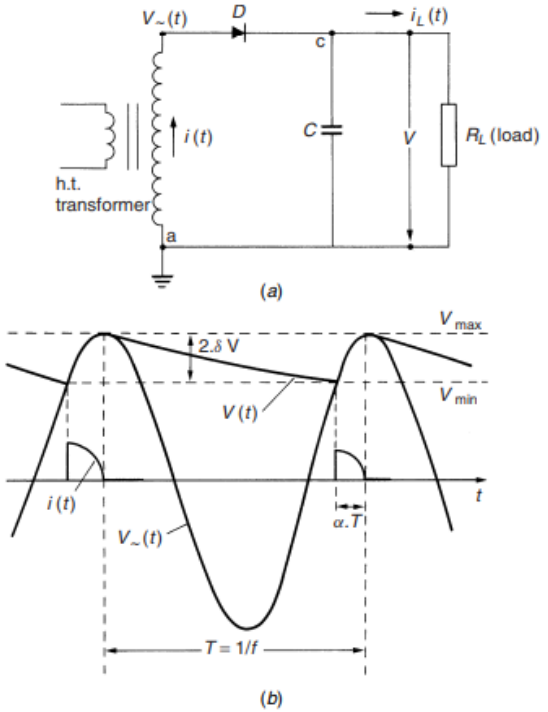


Figure 2.1 *Single-phase half-wave rectifier with reservoir capacitance C. (a) Circuit. (b) Voltages and currents with load R_L*

DIRECT VOLTAGES

The ripple δV could be calculated for this circuit based on the exponential decay of $V(t)$ during the discharge period $T(1-\alpha)$. However, for practical circuits, the neglected voltage drops within transformer and rectifiers must be taken into account. We may assume that $\alpha = 0$. Then δV is easily found from the charge Q transferred to the load,

$$Q = 2\delta VC = IT \qquad \delta V = \frac{IT}{2C} = \frac{I}{2fC}$$

This relation shows the interaction between the ripple, the load current and circuit parameter design values f and C. The mean output voltage will also be influenced by δV, even with a constant AC voltage and a lossless rectifier, no load-independent output voltage can be reached. The product of fC is an important design factor.

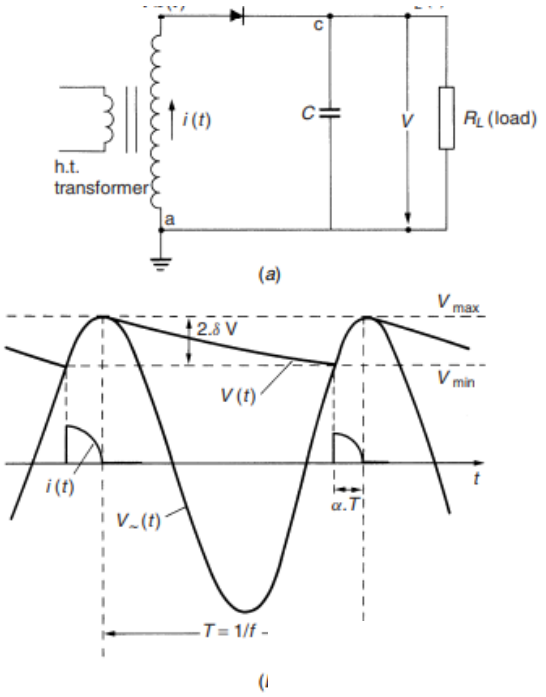


Figure 2.1 *Single-phase*
(a) Circuit. (b) Voltages an

2 DIRECT VOLTAGES

For HV test circuits, a sudden voltage breakdown at the load ($R_L \rightarrow 0$) must always be taken into account.

Whenever possible, the rectifiers should be able to carry either the excessive currents, which can be limited by fast, electroncailly controlled switching devices at the transformer input, or they ca nbe protected by an additional resistance inserted in the HV circuit. The latter method, however, increases the internal voltage drop.

Half-wave rectifier circuits have been built up to voltage in the megavolt range, in general by extending HV testing trasnformer to a DC current supply.

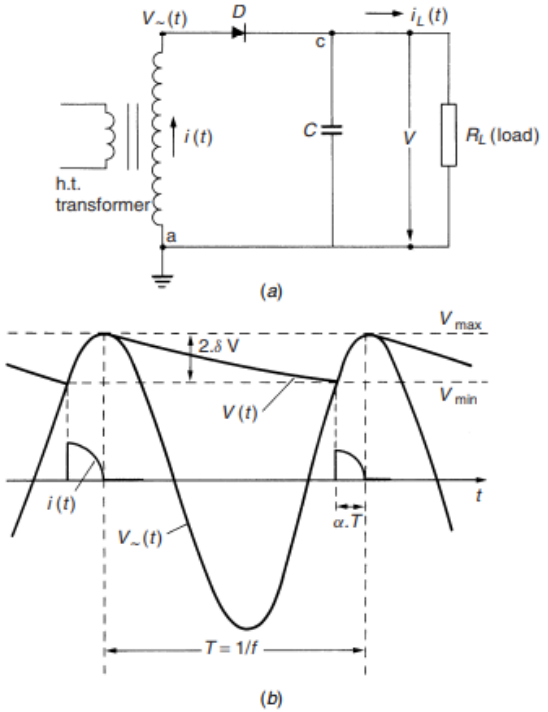


Figure 2.1 *Single-phase half-wave rectifier with reservoir capacitance C. (a) Circuit. (b) Voltages and currents with load R_L*

2 DIRECT VOLTAGES

A disadvantge of the single-phase half-wave rectifier is large size for pure DC supplies.

The other disadvantage is the possible saturation of the HV transfomer, if the amplitude of the DC is comparable with the nominal alternating current of the transformer.

The biphase half-wave (or single-phase full-wave) rectifier shown in the figure overcomes this disadvantage, but it does not change the fundamental efficiency, considering that two HV windings of the transformer are now available.

With reference to the frequency f during one cycle, now each of the diodes D_1 and D_2 is conducting for one half-cycle with a time delay of $T/2$. The ripple factor is halved. The real will be increased if both voltages $V_{1\sim}$ and $V_{2\sim}$ are not exactly equal. If V_{2max} would be smaller than $(V_{1max} - 2\delta V)$ or V_{min} this HV winding would not charge the capacitance C.

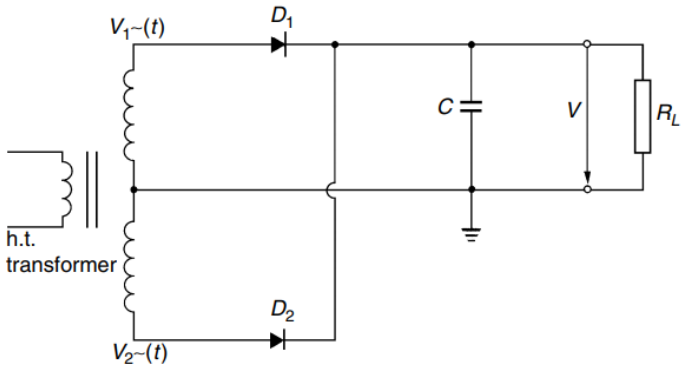


Figure 2.2 *Biphase half-wave rectifier circuit with smoothing capacitor C*

2 DIRECT VOLTAGES

Thus, single-phase full-wave circuits can only be used for HV applications if the HV windng of the transfomer can be earthed at its midpoint and if the DC output is single-ended grounded.

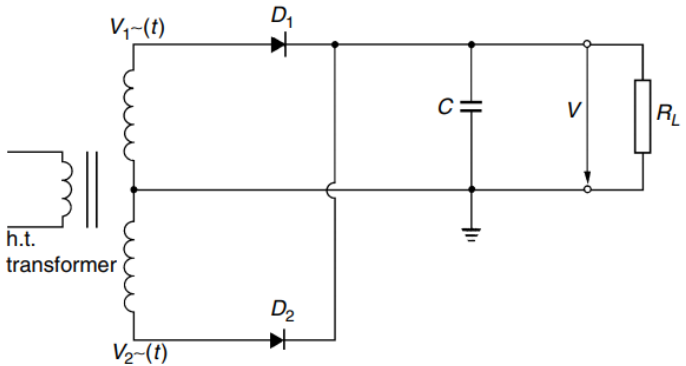


Figure 2.2 *Biphase half-wave rectifier circuit with smoothing capacitor C*

Cascaded circuits

The demand for very high voltages forced the improvement of rectifying circuits. In a cascaded circuit, in every multiplier circuit, transformers, rectifiers and capacitor units have only to withstand a fraction of the total output voltage.

An n-stage single-phase cascaded circuit is shown in the figure.

When HV output is open-circuited: $I = 0$. The portion $0 - n' - V(t)$ is a half-wave rectifier circuit in which C'_n charges up to a voltage of $+V_{max}$ if $V(t)$ has reached the lowest potential, $-V_{max}$. If C_n is still uncharged, the rectifier D_n conducts as soon as $V(t)$ increases.

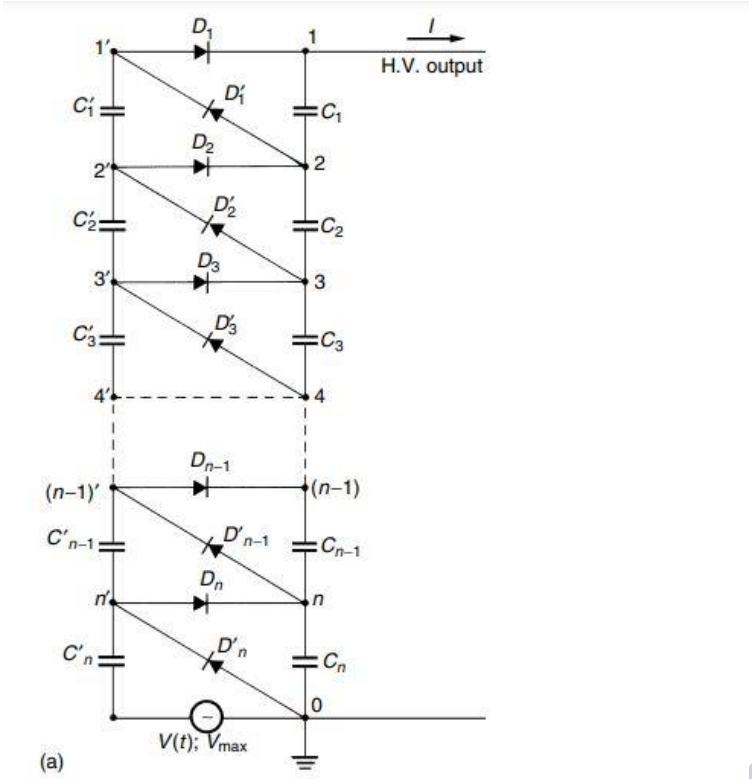


Figure 2.3 *(a) Cascade circuit according to Cockroft–Walton or Greinacher. (b) Waveform of potentials at the nodes, no load*

Cascaded circuits

As the potential of point n' swings up to $+V_{2max}$ during the period $T = 1/f$, point n attains further on a steady potential of $+2V_{max}$ if $V(t)$ has reached the highest potential of $+V_{max}$.
The part $n' - n - 0$ is therefore a half-wave rectifier, in which the voltage across D'_n can be assumed to be the AC voltage source.

The current through D_n that charged the C_n was not provided by D'_n, but from $V(t)$ and C'_n. We assumed, therefore that C'_n was not discharged, which is not correct. As we will consider for the loaded circuit, we can also assume that the voltage across C_n is not reduced if the potential of n', osciallates between zero and $+2V_{max}$.

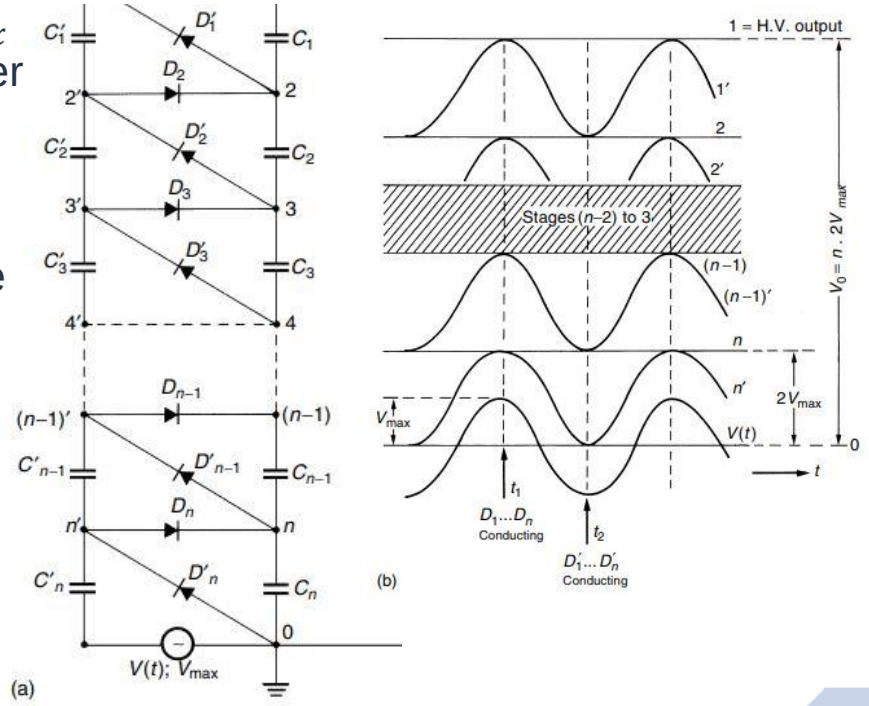


Figure 2.3 *(a) Cascade circuit acc Greinacher. (b) Waveform of potentia*

Cascaded circuits

If the potential of n', is zero, the capacitor C'_{n-1} is also charged to the potential of n,i.e. to a voltage of $+2V_{max}$. The next voltage oscillation of $V(t)$ from $-V_{max}$ to $+V_{max}$ will force the diode D_{n-1} to conduct, so that also C_{n-1} will be charged to a voltage of $+2V_{max}$.

From the sketched signals in the figure:

- the potentials at the nodes $1', 2' \ldots n'$ are oscillating due to the voltage oscillation of $V(t)$,

- the potentials at the nodes $1, 2 \ldots n$ remain constant with reneference to ground potential,

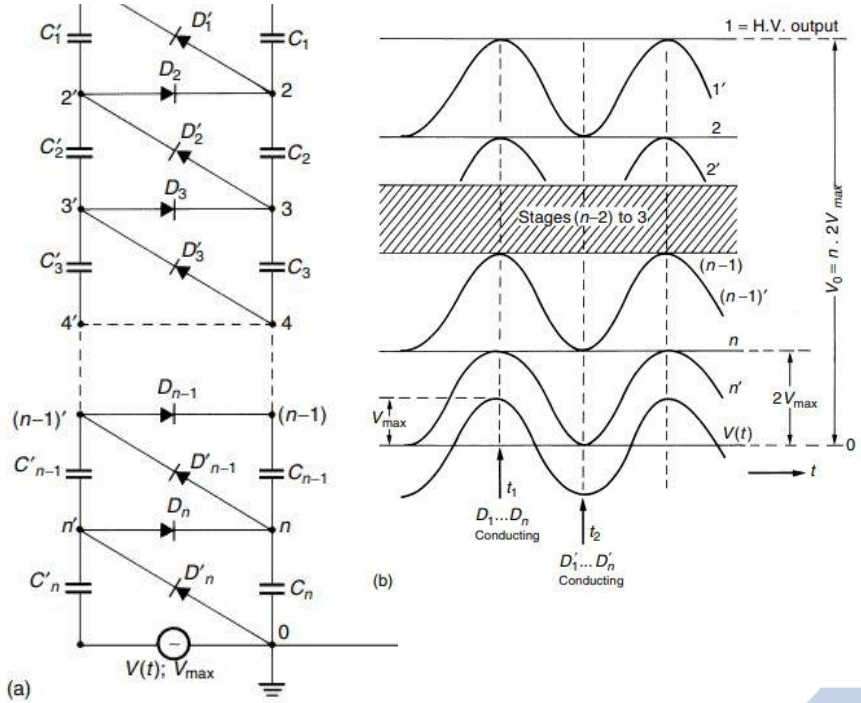


Figure 2.3 *(a) Cascade circuit acc*
Greinacher. (b) Waveform of potentia

Cascaded circuits

- the voltages across all capacitors are DC type with $2V_{max}$ magnitude across each capacitor stage, except the capacitor C'_n which is stressed with V_{max} only,

- every rectifier $D_1, D'_1 \dots D_n, D'_n$ is stressed with $2V_{max}$ or twice AC peak voltage,

- the HV output will reach a maximum voltage of $2nV_{max}$.

Use of several stages in this manner enables very high voltages to be obtained.
The equal stress of the elements used allows a modular design of such generators.
The number of stages is strongly limited by the current due to any load.

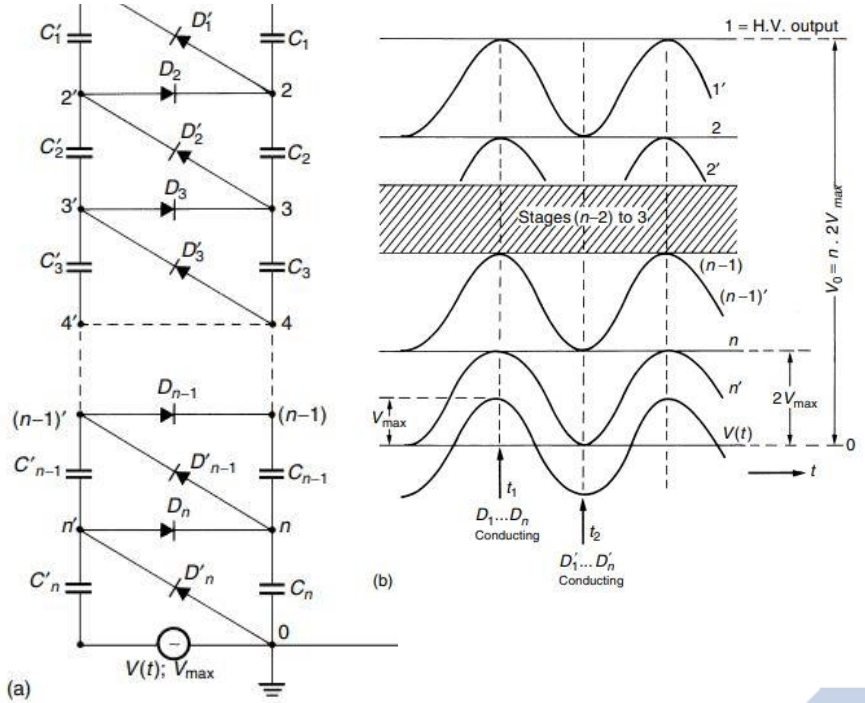


Figure 2.3 *(a) Cascade circuit acc Greinacher. (b) Waveform of potentia*

Cascaded circuits

When HV output loaded: $I > 0$. If the generator supplies any load current I, the output voltage will never reach the value $2nV_{max}$ as shown in the figure left. There will also be a ripple on the voltage, and therefore we have to deal with two quantities, the voltage drop ΔV_0 and the peak-to-peak ripple $2\delta V$. The figure below shows the shape of the output voltage, ΔV_0 and $2\delta V$.

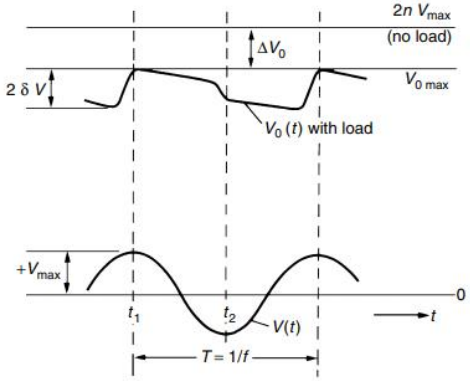
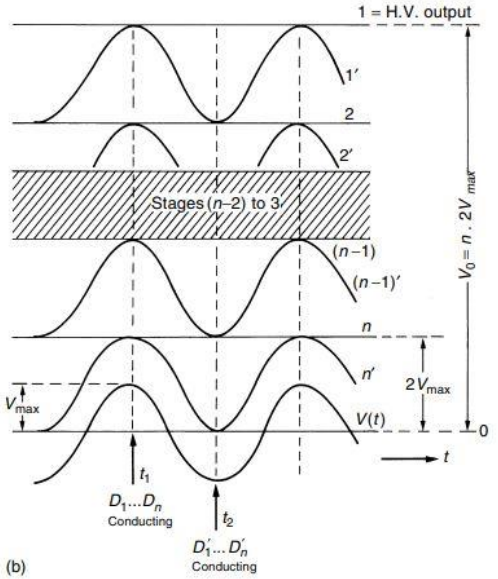


Figure 2.4 *Loaded cascade circuit, definitions of voltage drop ΔV_0 and ripple δV*



Cascaded circuits

